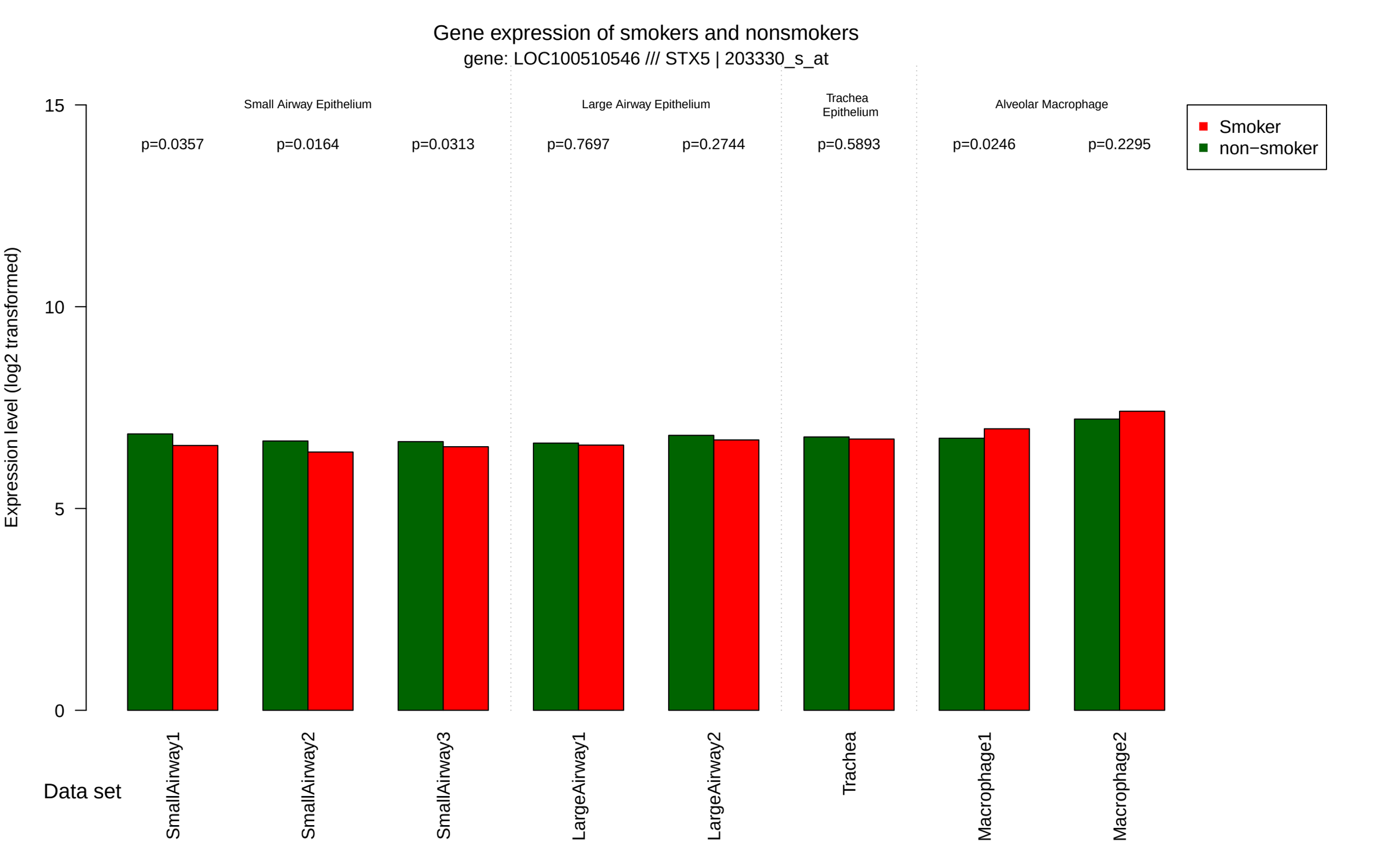
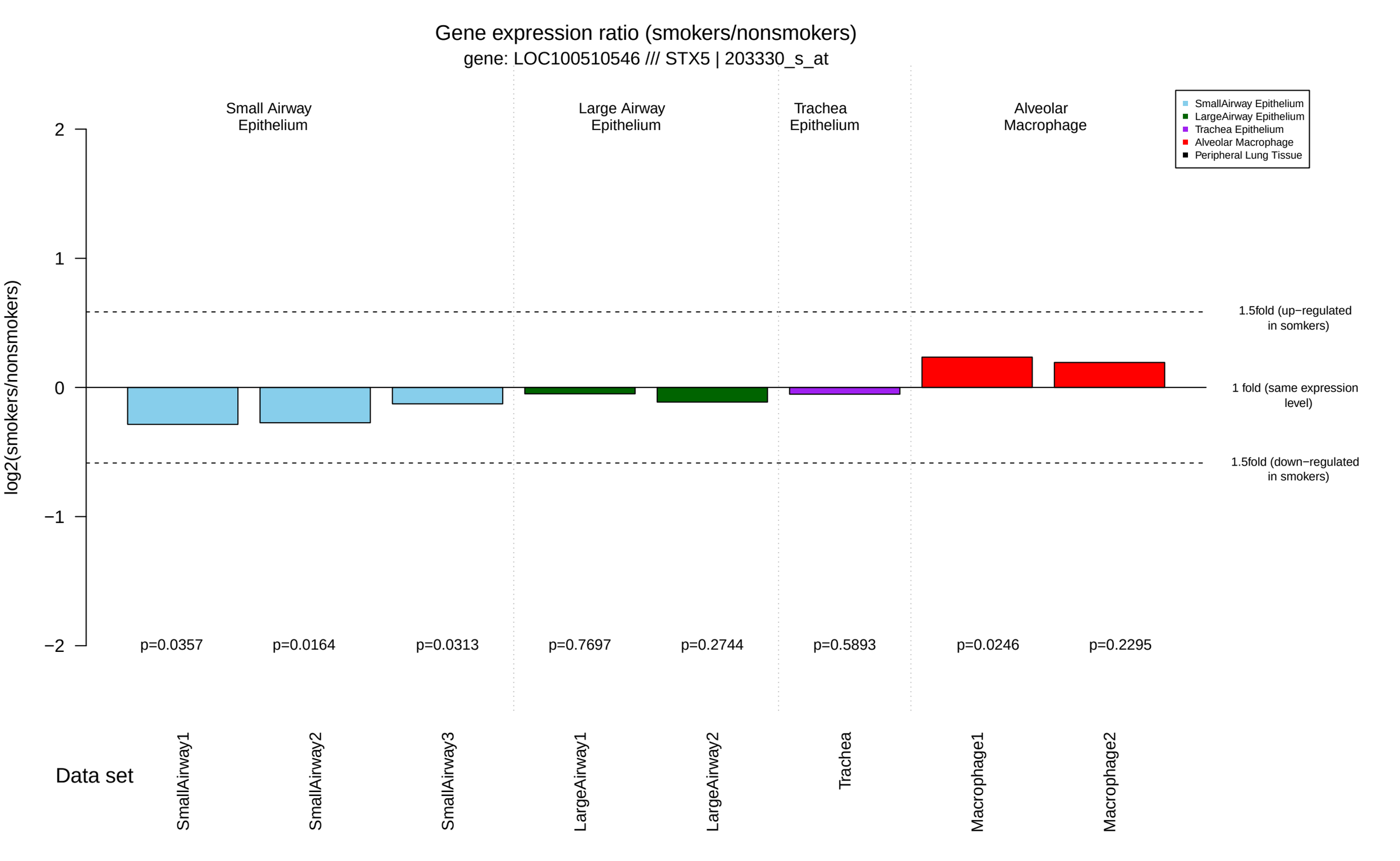
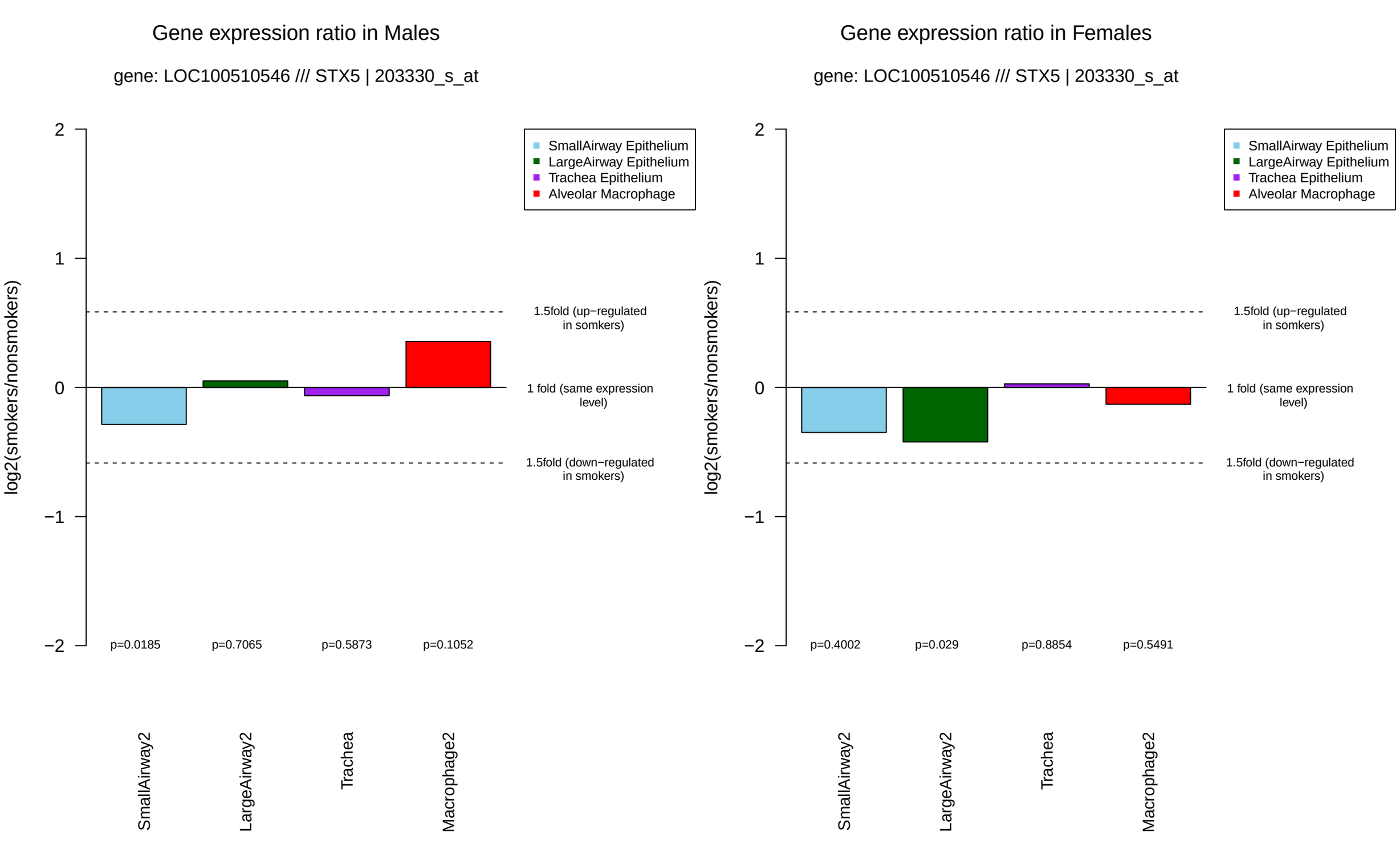
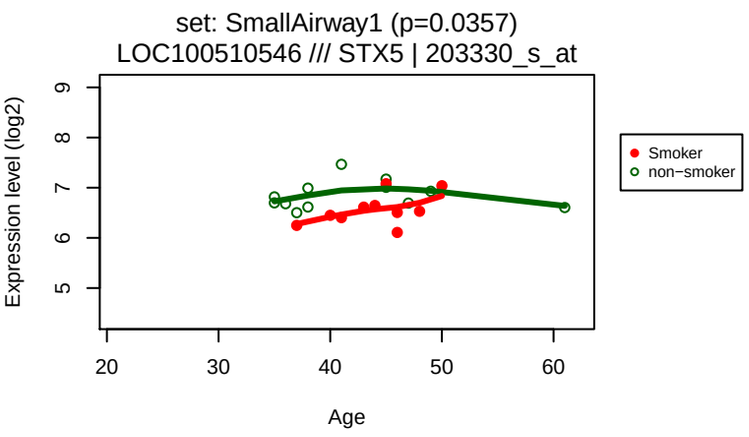
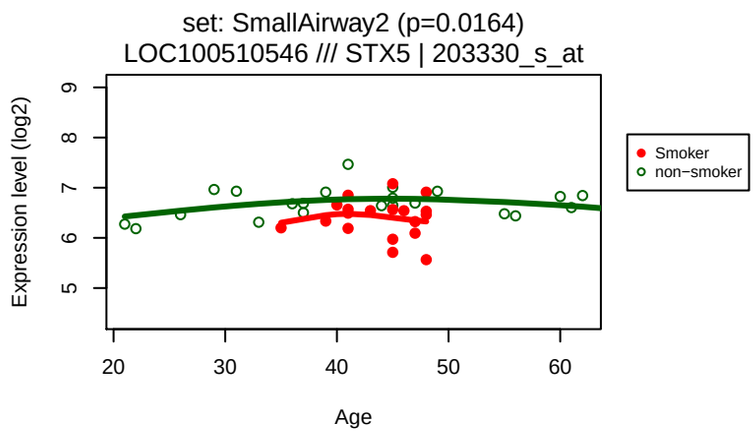
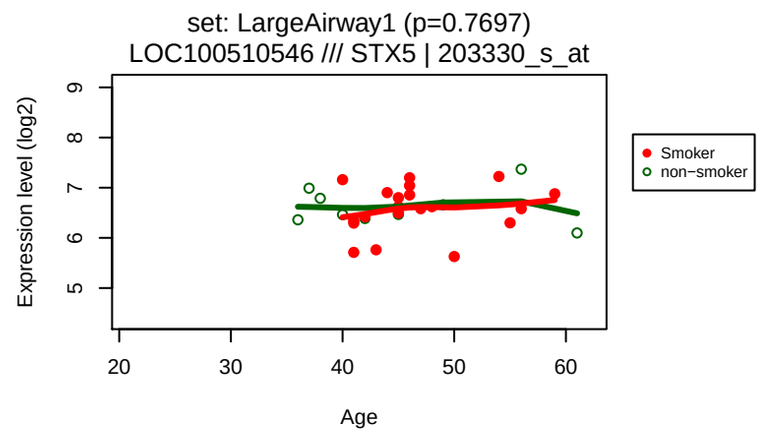

Gene expression of smokers and nonsmokers
gene: LOC100510546 /// STX5 | 203330_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0357
p=0.0164
p=0.0313
p=0.7697
p=0.2744
p=0.5893
p=0.0246
p=0.2295
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

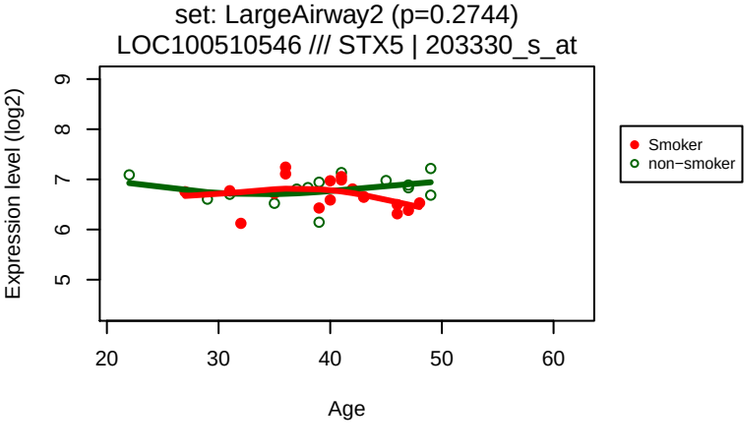

Gene expression ratio (smokers/nonsmokers)
gene: LOC100510546 /// STX5 | 203330_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0357
p=0.0164
p=0.0313
p=0.7697
p=0.2744
p=0.5893
p=0.0246
p=0.2295
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

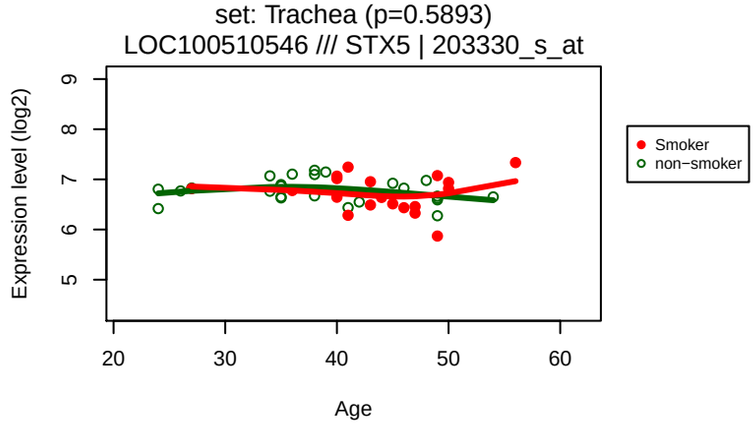

Gene expression ratio in Males
gene: LOC100510546 /// STX5 | 203330_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0185
p=0.7065
p=0.5873
p=0.1052
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100510546 /// STX5 | 203330_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4002
p=0.029
p=0.8854
p=0.5491
SmallAirway2
LargeAirway2
Trachea
Macrophage2

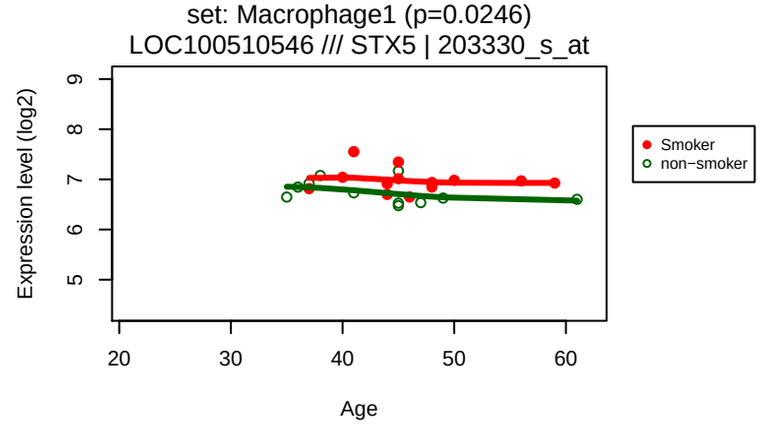

set: SmallAirway1 (p=0.0357)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker

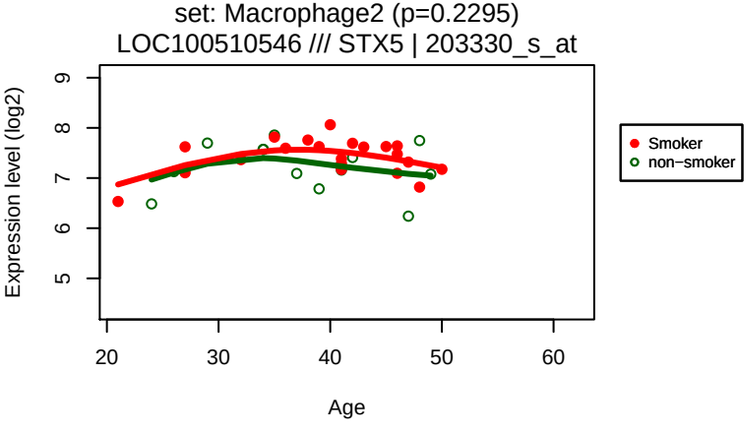

set: SmallAirway2 (p=0.0164)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker

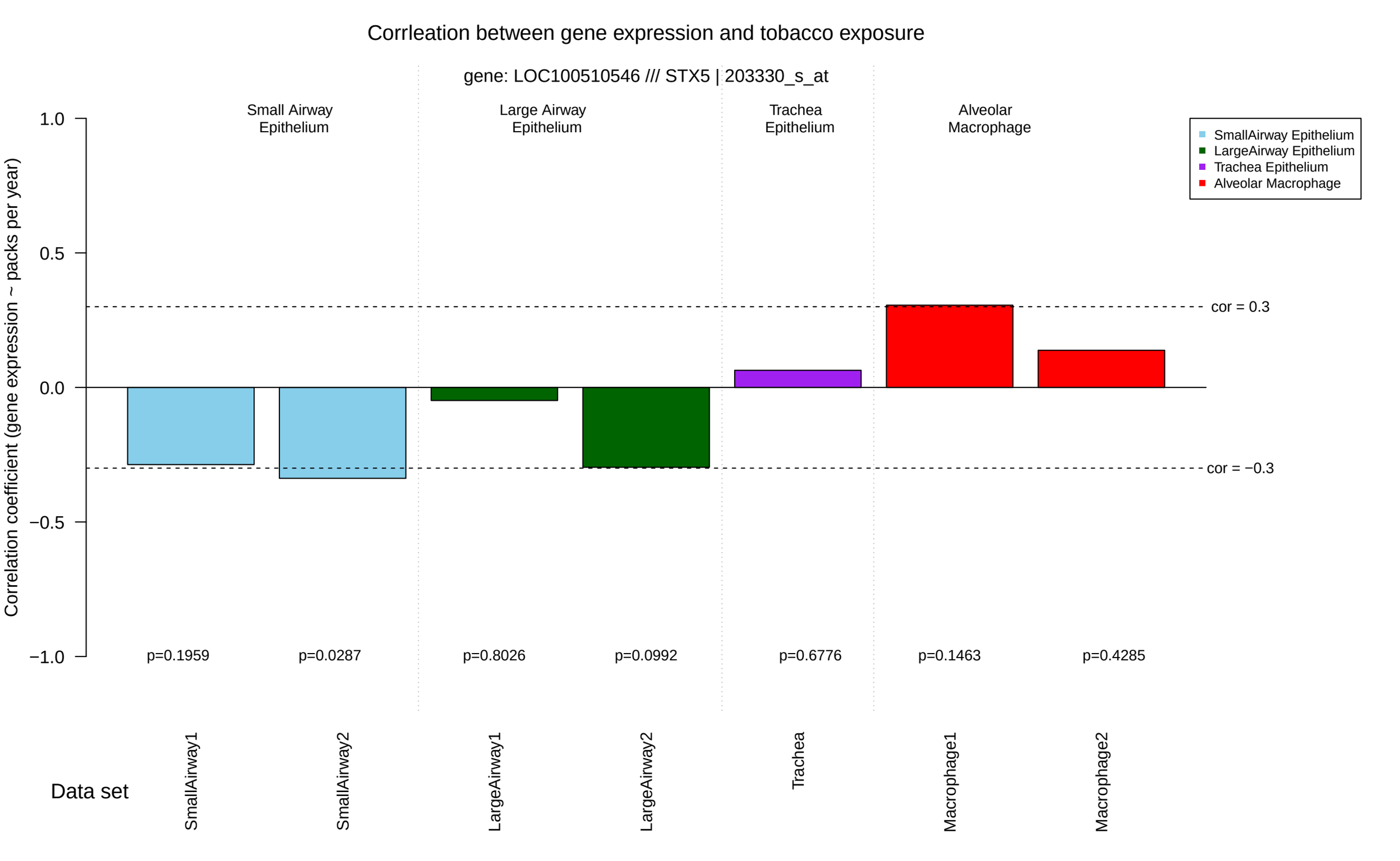

set: LargeAirway1 (p=0.7697)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2744)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5893)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0246)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2295)
LOC100510546 /// STX5 | 203330_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100510546 /// STX5 | 203330_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1959
p=0.0287
p=0.8026
p=0.0992
p=0.6776
p=0.1463
p=0.4285
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2